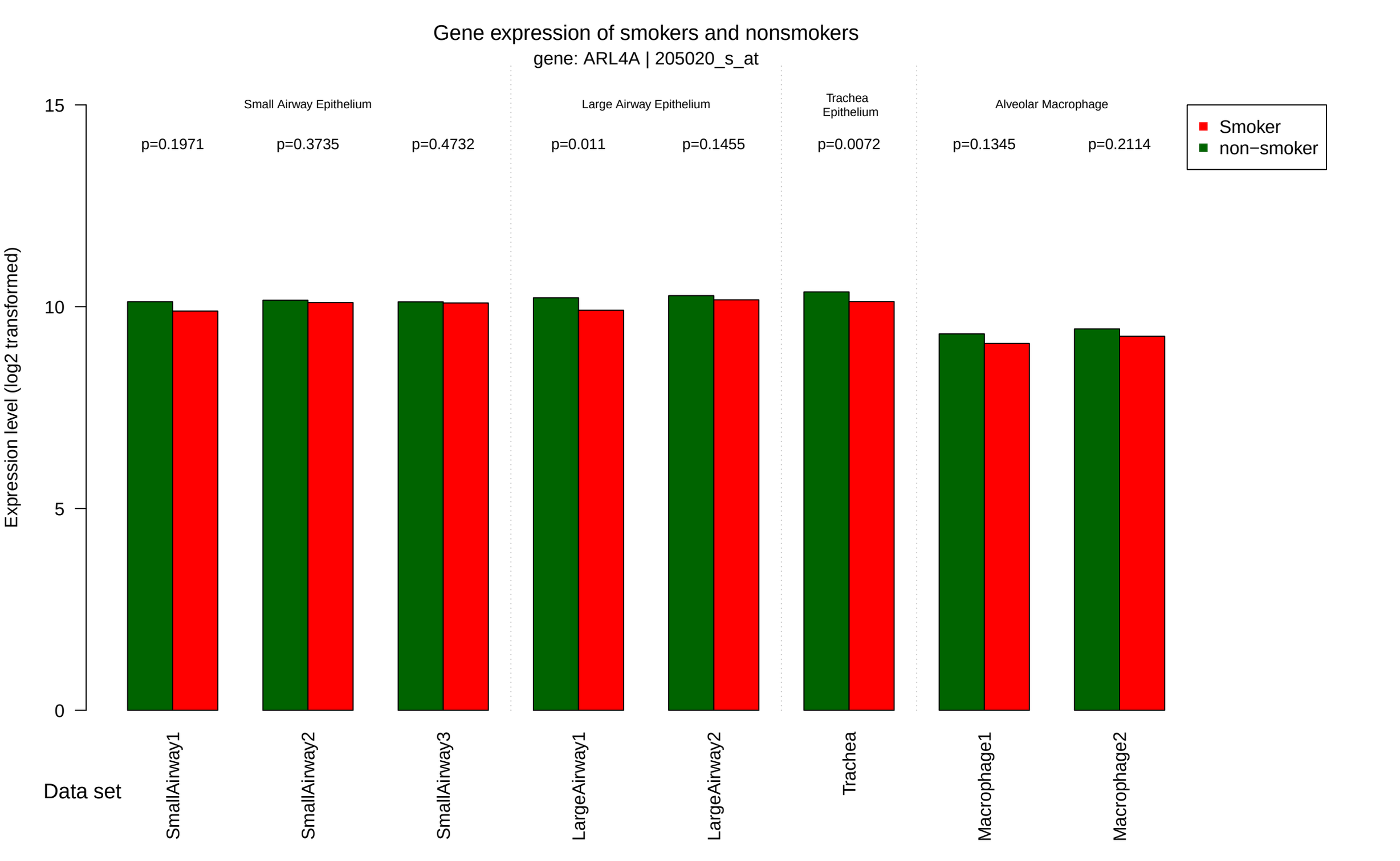
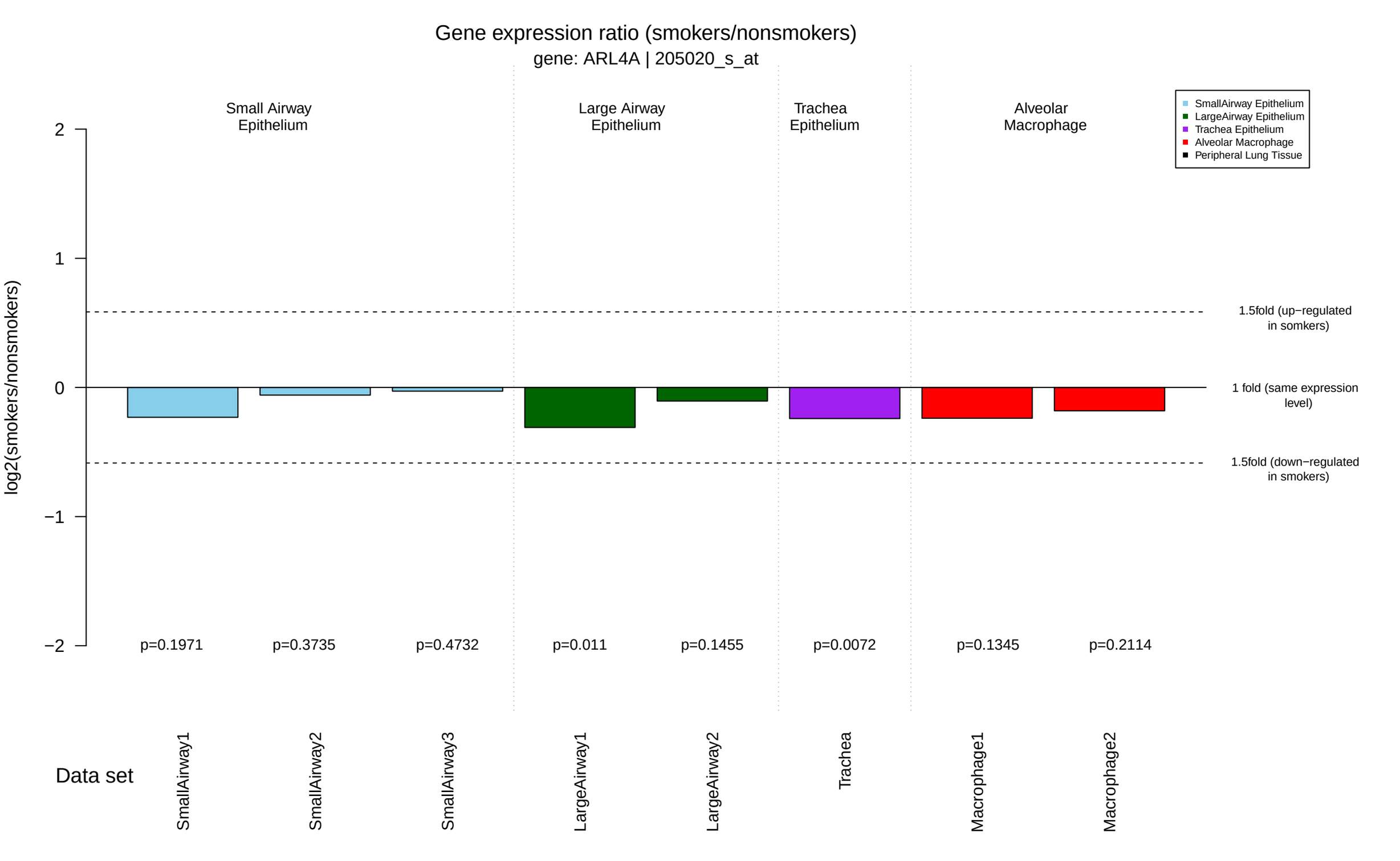
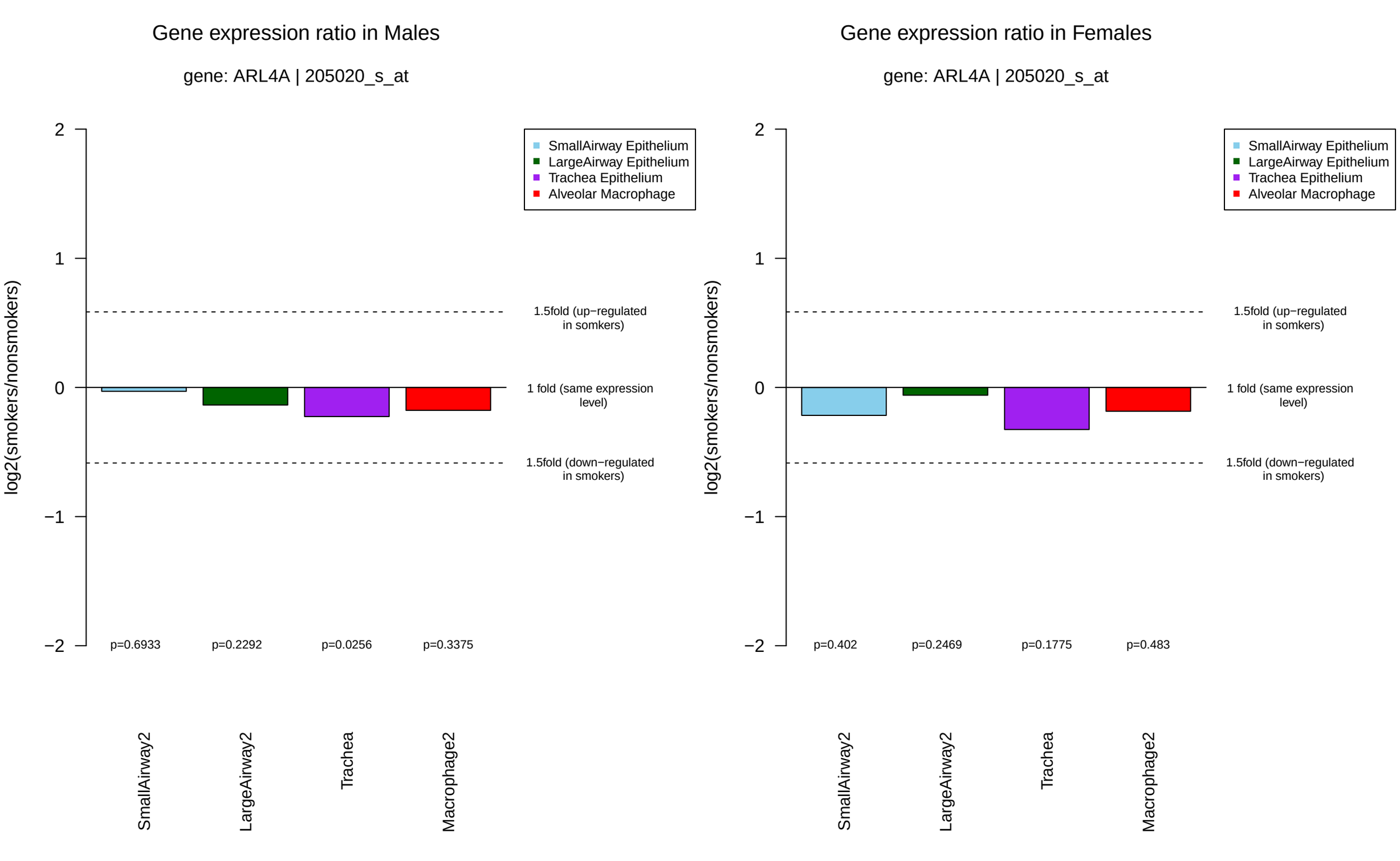
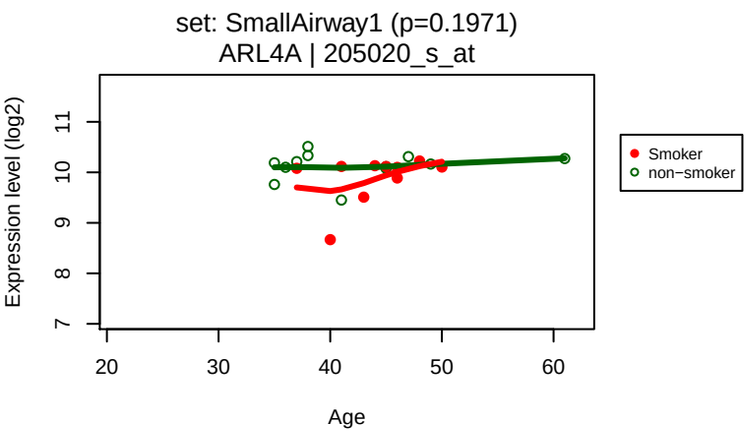
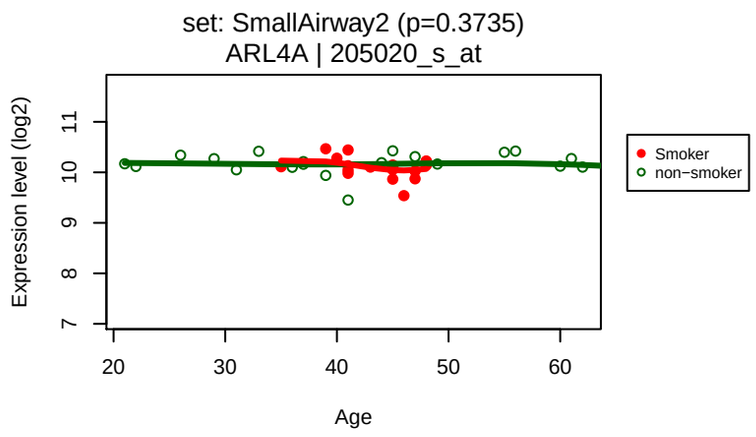
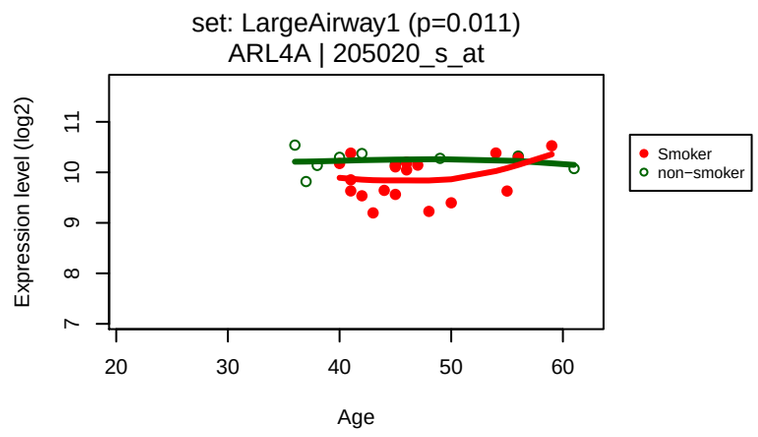

Gene expression of smokers and nonsmokers
gene: ARL4A | 205020_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1971
p=0.3735
p=0.4732
p=0.011
p=0.1455
p=0.0072
p=0.1345
p=0.2114
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

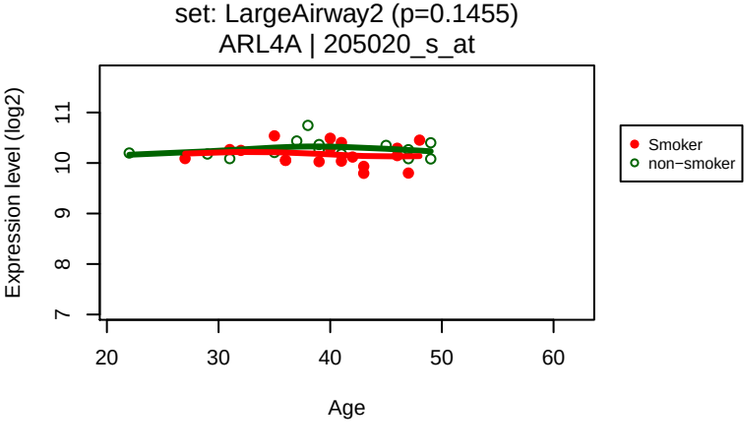

Gene expression ratio (smokers/nonsmokers)
gene: ARL4A | 205020_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1971
p=0.3735
p=0.4732
p=0.011
p=0.1455
p=0.0072
p=0.1345
p=0.2114
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

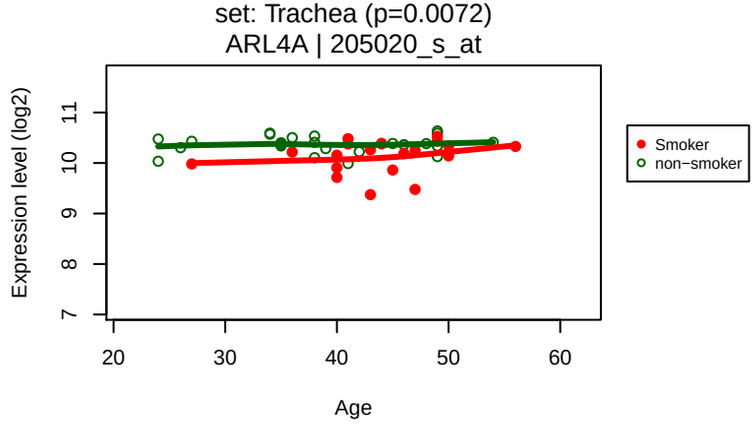

Gene expression ratio in Males
gene: ARL4A | 205020_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6933
p=0.2292
p=0.0256
p=0.3375
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL4A | 205020_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.402
p=0.2469
p=0.1775
p=0.483
SmallAirway2
LargeAirway2
Trachea
Macrophage2

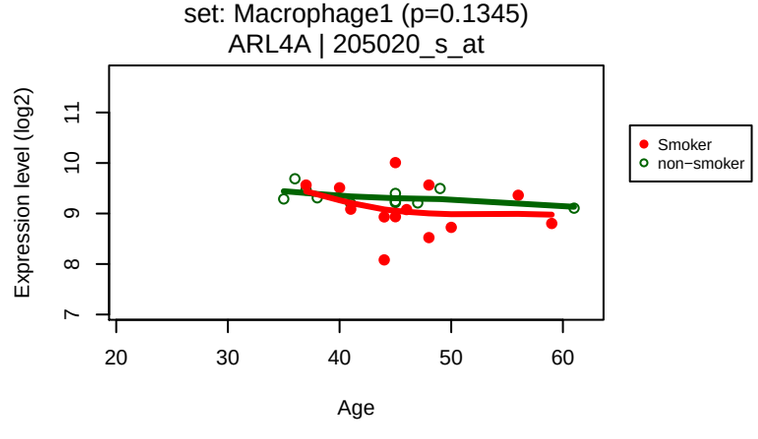

set: SmallAirway1 (p=0.1971)
ARL4A | 205020_s_at
Expression level (log2)
Age
Smoker
non−smoker

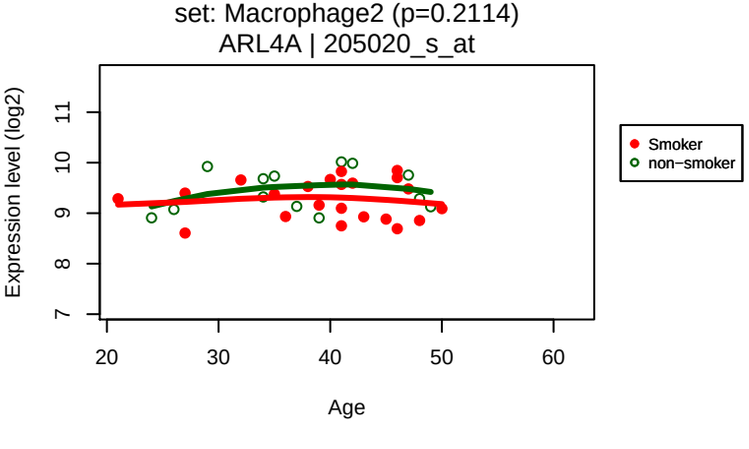

set: SmallAirway2 (p=0.3735)
ARL4A | 205020_s_at
Expression level (log2)
Age
Smoker
non−smoker

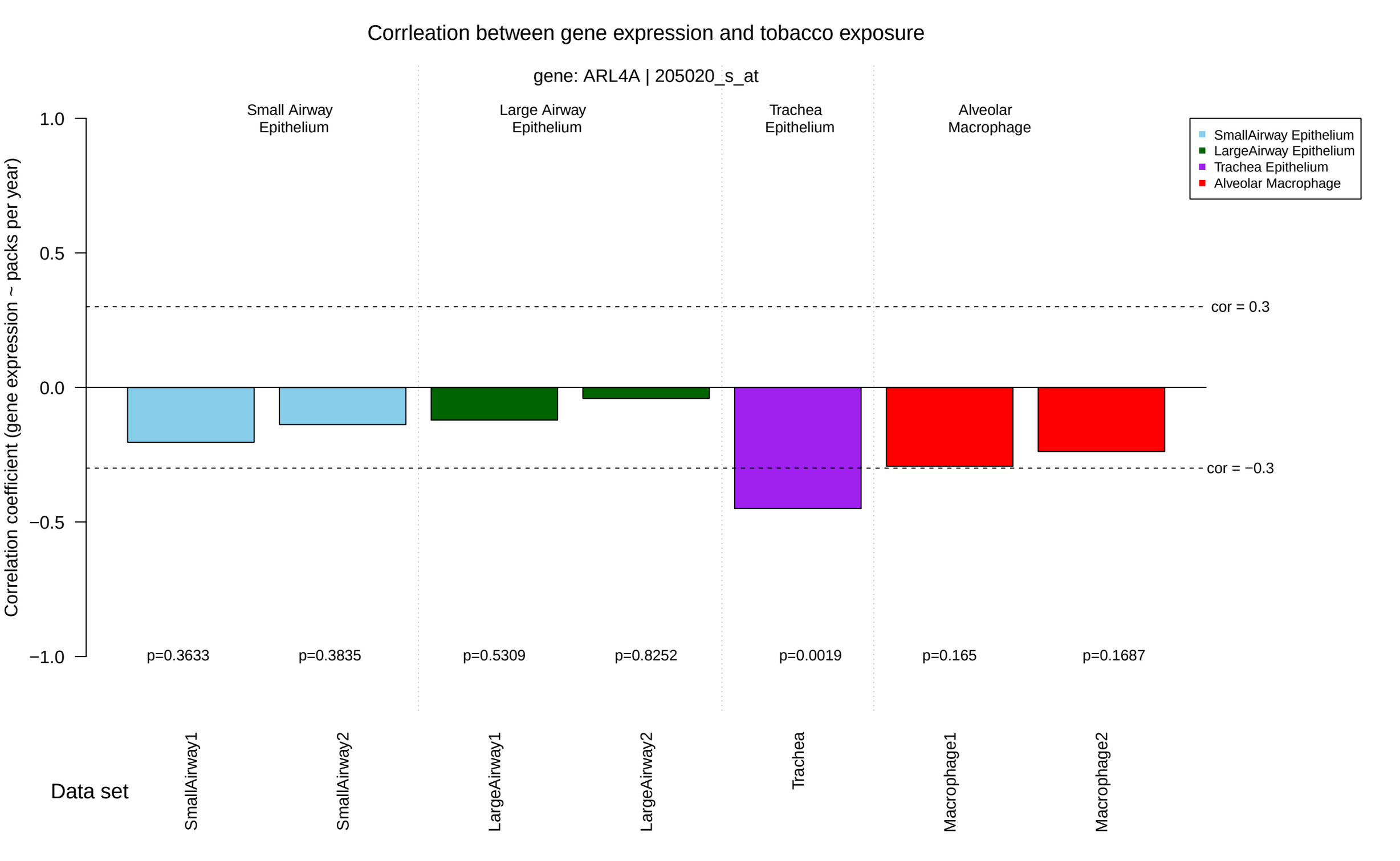

set: LargeAirway1 (p=0.011)
ARL4A | 205020_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1455)
ARL4A | 205020_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0072)
ARL4A | 205020_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1345)
ARL4A | 205020_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2114)
ARL4A | 205020_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL4A | 205020_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3633
p=0.3835
p=0.5309
p=0.8252
p=0.0019
p=0.165
p=0.1687
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2